

EDUCATION

University of North Texas

Masters in. Data Science | GPA:3.83/4.0

Jan 2023 – Dec 2024

Relevant Courses: Machine Learning, Deep Learning, NLP, SE for data scientists, Fundamentals of Data Science, Data Security, Data Mining, Data Processing, Data visualization, Virtual Reality Applications.

SUMMARY

- 5 years of experience in data handling and machine learning solutions, with a concentration on Azure Data Factory (ADF), Amazon Redshift and SQL.
- Directed data pipeline monitoring and optimization in Azure Data Factory to ensure seamless data flow and prompt processing to meet business requirements.
- Managed data partitioning in Amazon Redshift, improving query performance, storage space usage, and system responsiveness as a whole.
- Troubleshoot and fixed job failures using reverse engineering techniques, identifying the root cause and implementing long-term solutions for enhancing job reliability.
- Searched and fixed ODBC and JDBC exceptions for seamless data connectivity between platforms and databases.
- Applied regular expressions (Regex) for data validation, transformation, and cleansing for improved data accuracy and integrity through ETL pipelines.
- Handled and closed ServiceNow tickets for job execution failure with zero downtime and real-time issue resolution.
- Conducted root cause analysis (RCA) of recurring job failures and initiated corrective action to avoid future failures.
- Created weekly customer reports, detailing job execution issues, debugging status, and recommendations for performance improvement.
- Demonstrated leadership abilities by leading team members, sharing best practices in debugging and data optimization.
- Collaborated with cross-functional teams such as data engineers, data analysts, and business stakeholders to enhance ETL processes and data processing optimization.
- Trained machine learning applications, with data engineering solutions present to enable deep analytics, model training, and deployment.

TECHNICAL SKILLS

ProgrammingPython, Java, JavaScript, C++, SQL

AI LibrariesNumpy, scipy, pandas, matplotlib, seaborn, Sklearn, OpenCV, Tensorflow, PyTorch, Beautiful Soup, cv2, TensorRT

Database & CloudMySQL, PostgreSQL, GitHub, MongoDB, AWS Redshift, BigQuery

DomainsMachine Learning, CNNs, Deep Learning, Time series Forecasting, Data Engineering, Data manipulation, Databricks, PySpark, Hadoop, Hive, Spark, Scala, Sqoop, Kafka, Kubernetes, Tableau, PowerBI, Elastic Map Reduce(EMR), Resilient Distributed Systems, AWS Quicksight, VS code, Analytical Skills, LLMs, GenAI, Data Validation, Data Analytics, Data Visualization, Data Wrangling, Cloud Computing, Cloud engineering, Cloud security, Data storage, GCP Compute Engine, Cloud SQL, Cloud Dataproc, Cloud SQL, Cloud dataflow, Cloud ML.

WORK EXPERIENCE

NIKE India Pvt Ltd (COGNIZANT (AI &A))

Jr. Data Scientist

Bengaluru, India

Dec 2021- Dec 2022

- Built and deployed machine learning models for retail analytics like sales forecast, demand forecast, and customer segmentation.
- Created and optimized ETL pipelines using AWS platforms (S3, Redshift, Athena, Glue) to manage large-sized retail data.
- Conducted exploratory data analysis (EDA) of customer behavior, inventory trends, and price strategies to drive business decisions.
- Built and tuned predictive models for demand forecasting, churn prediction, and product recommendation using Python, Scikit-Learn, and TensorFlow.
- Automated data reporting and processing business processes on AWS Step Functions, Lambda, and SQL to improve operational efficiency.
- Created interactive dashboards and reports on Tableau, QuickSight, and Matplotlib to display sales performance and inventory optimization information.
- Deployed customer segmentation models through clustering algorithms (K-Means, DBSCAN) to better personalize marketing campaigns.
- Worked with big data technologies (PySpark, Pandas, NumPy) to process and analyze transactional and consumer data at scale.
- Collaborated with business stakeholders to transform data insights into effective retail strategies that drive sales, inventory management, and customer engagement.
- Managed and optimized data partitioning in Amazon Redshift, improving query performance and storage efficiency.
- Ensured model scalability and deployment using AWS SageMaker and MLOps best practices for continuous model tuning.

Retail Business Services (RBS) (COGNIZANT (AI &A))

Data Engineer

Chennai, India

Aug 2020- May 2021

- Tracked and streamlined Azure Data Factory (ADF) pipelines to ensure data workflow execution on time to meet ETA standards.
- Managed and corrected job failure by reviewing and responding to ServiceNow tickets to ensure zero downtime and business continuity.
- Applied reverse engineering techniques for debugging and troubleshooting data pipeline failure for improved job reliability and performance.
- Investigated and corrected ODBC and JDBC errors to facilitate seamless data connectivity and integration among data sources.
- Used applied regular expressions (Regex) to verify and reformat data, giving accuracy and consistency in data processing.
- Conducted root cause analysis (RCA) on standard job failures and implemented long-term solutions to prevent future occurrences.
- Collaborates with cross-functional teams of developers and business analysts to maximize ETL processes and data processing performance.
- Submitted weekly reports to the client, appraising them of job run-time issues, fixes, and process improvement recommendations.
- Enabled compliance with data governance policies in handling mass-retail data, improving data integrity and business process effectiveness.

- I trained and worked in Bigdata tools like Apache Hadoop, Hive, Spark, Scala and designed DBMS using MySQL and languages like Java.
- Worked on creating internal and external tables using Hive and used various methods like bucketing and partitioning of the data for better understanding.
- Experienced on handling raw data processing using data cleaning and data validation techniques in big data tools.

PROJECTS

Social Bias Detection using Multimodal Embeddings

- Developed a system to detect social bias (such as gender, race, or age bias) in text data by leveraging the contextual understanding of BERT pre-trained LLM models.
- Implemented a context-aware approach to accurately identify and mitigate implicit biases in online content, such as social media posts, news articles, and comments.

Tumor Classification Using Machine Learning Algorithms

- Optimized a Convolutional Neural Network with three hidden layers, achieving a 95% accuracy rate in classifying tumors from a lung cancer image dataset.
- Conducted comprehensive testing on baseline, probabilistic, and Random Forest models, ultimately selecting the optimal model for precise and reliable predictions.

Spotify Playlist Generation – Unsupervised Learning

- Implementing DBScan, KNNs to enhance feature extraction from diverse datasets, contributing to the development of innovative solutions and demonstrating a proficiency in data-driven decision-making.
- Leveraged advanced data analysis techniques, including K-means clustering, on a Spotify dataset to identify the optimal number of user preference clusters, leading to improved insights and a more personalized user experience.

Air Quality Index (AQI) Prediction and Analysis using Amazon Web Services (AWS) tools.

- Developed AWS-based AQI Prediction and Analysis solution, optimizing forecasting accuracy and risk mitigation for weather events. Acted as both developer and tester, ensuring project reliability across AWS CLI, S3, EMR, Redshift, Jupyter and Quicksight.